When can magic be easy to simulate?

Efficiently benchmarking quantum error correction

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merlin-sim

Joint work with



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Oxford

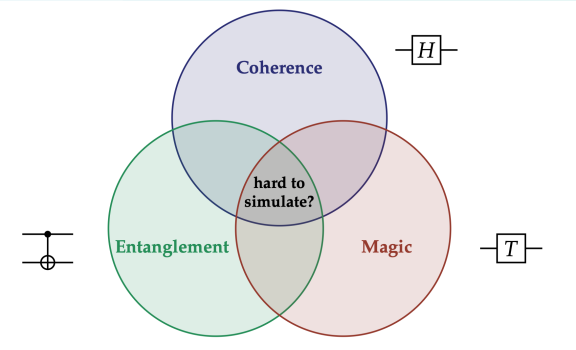


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Oxford

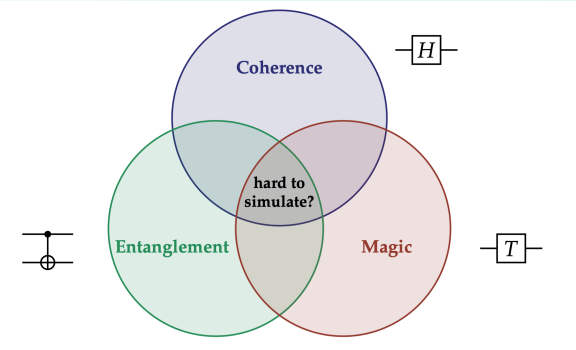
arXiv

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What quantum circuits are classically simulable?

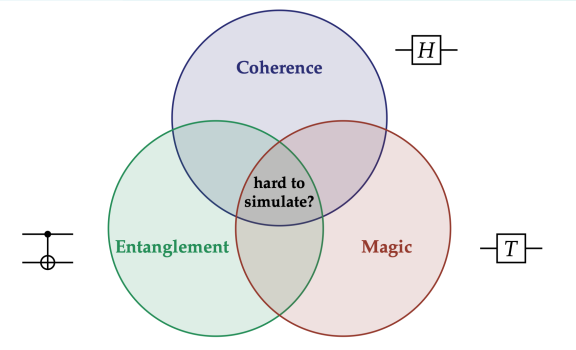


What quantum circuits are classically simulable?



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The answer has practical consequences for **quantum error correction**.

Quantum error correction

The idea: add redundancy to protect information against errors.

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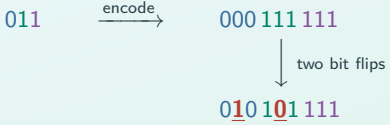
Classical example: repeat each bit three times.

$$011 \xrightarrow{\text{encode}} 000\,111\,111$$

Quantum error correction

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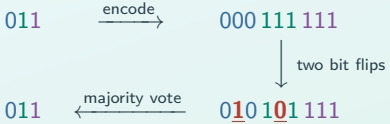
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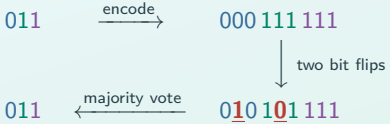
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Quantum version: encode each logical qubit across several physical qubits.

Measure error checks without reading out the stored information.

Gates on encoded qubits

Each gate in the algorithm needs a **fault-tolerant logical implementation**.

Algorithm



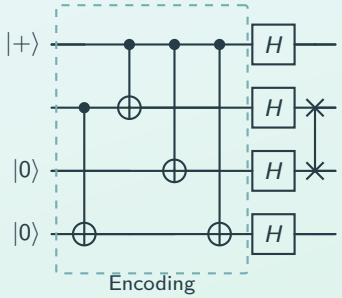
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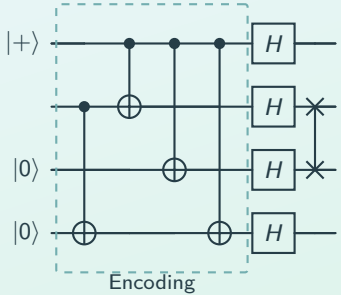
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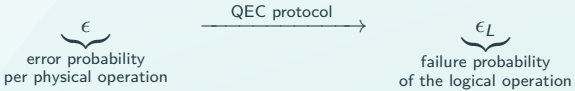
Encoded implementation



A small circuit becomes a **larger circuit on more qubits**.

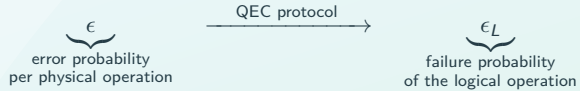
Benchmarking quantum error correction

A protocol converts physical errors into a logical error rate:



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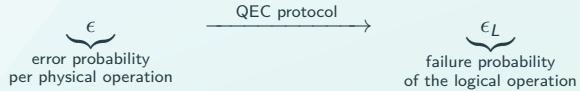
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Benchmarking quantum error correction

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How? Simulate noisy protocols, count logical failures, and compare designs.

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Can we exploit the structure of quantum error correction?

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If magic alone is not enough, what makes quantum circuits hard to simulate?

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